

Effect of a very-high-fiber vegetable, fruit, and nut diet on serum lipids and colonic function.



We tested the effects of feeding a diet very high in fiber from fruit and vegetables. The levels fed were those, which had originally inspired the dietary fiber hypothesis related to colon cancer and heart disease prevention and also may have been eaten early in human evolution. Ten healthy volunteers each took 3 metabolic diets of 2 weeks duration. The diets were: high-vegetable, fruit, and nut (very-high-fiber, 55 g/1,000 kcal); starch-based containing cereals and legumes (early agricultural diet); or low-fat (contemporary therapeutic diet). All diets were intended to be weight-maintaining (mean intake, 2,577 kcal/d). Compared with the starch-based and low-fat diets, the high-fiber vegetable diet resulted in the largest reduction in low-density lipoprotein (LDL) cholesterol (33% $\pm$ 4%, $P < .001$) and the greatest fecal bile acid output (1.13 $\pm$ 0.30 g/d, $P = .002$), fecal bulk (906 $\pm$ 130 g/d, $P < .001$), and fecal short-chain fatty acid outputs (78 $\pm$ 13 mmol/d, $P < .001$). Nevertheless, due to the increase in fecal bulk, the actual concentrations of fecal bile acids were lowest on the vegetable diet (1.2 mg/g wet weight, $P = .002$). Maximum lipid reductions occurred within 1 week. Urinary mevalonic acid excretion increased ($P = .036$) on the high-vegetable diet reflecting large fecal steroid losses. We conclude that very high-vegetable fiber intakes reduce risk factors for cardiovascular disease and possibly colon cancer. Vegetable and fruit fibers therefore warrant further detailed investigation. Copyright 2001 by W.B. Saunders Company